CLASS 8 – CBSE MATHEMATICS QUESTION PAPER

Time Allowed: 1 Hour

SECTION A: SHORT ANSWER QUESTIONS

(4 questions × 2 marks = 8 Marks)

Q1. Exponents & Powers:

Using exponent laws, evaluate and express the final answer in single exponent form with a positive index:

$$\left(\left(\frac{2}{3}\right)^{-2} \times \left(\frac{3}{4}\right)^{-3}\right) \div \left(\frac{8}{9}\right)^{-2}$$

Q2. Algebraic Identities:

If $x + \frac{1}{x} = 6$, find the exact value of $x^2 + \frac{1}{x^2}$ using standard algebraic expansion formulas.

Q3. Direct and Inverse Variations:

A team of 12 workers can construct a boundary wall in 20 days working 8 hours a day. Using proportional inverse variation relationships, calculate how many days 16 workers will take to complete the same wall working 6 hours a day.

Q4. Factorisation:

Factorise the quadratic expression completely using the identity $a^2 - b^2 = (a - b)(a + b)$ or quadratic splitting:

$$16x^4 - 81y^4$$

SECTION B: MEDIUM ANSWER QUESTIONS

(4 questions × 3 marks = 12 Marks)

Q5. Mensuration (Trapezium & Trapezoidal Prisms):

The cross-section of a canal is a trapezium. The parallel sides of the canal cross-section are 12 m and 8 m wide, and the perpendicular depth between them is 4 m.

(a) Calculate the cross-sectional area of the canal using the trapezium area formula $A = \frac{1}{2}(a + b)h$. *(1.5 marks)*

(b) If the canal is 250 m long, calculate the total volume of earth dug out to build this canal section. *(1.5 marks)*

Q6. Comparing Quantities (Compound Interest & Depreciation Formulas):

A business purchased laboratory equipment worth Rs. $1,25,000$. Its value depreciates annually at the rate of $R_1 = 8\%$ during the first year and $R_2 = 10\%$ during the second year.

Using the compound interest formula $A = P\left(1 - \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)$:

(a) Calculate the net depreciated value of the equipment at the end of 2 years.

(b) Find the total drop in value (total depreciation amount).

Q7. Algebraic Expansion & Simplification:

Expand and simplify the combined expression using suitable algebraic identities $(a+b)^2$, $(a-b)^2$, and $(a+b)(a-b)$:

$$(3x+4y)^2-(3x-4y)^2+(3x-4y)(3x+4y)$$

Q8. Solid Geometry (Cylinder & Hollow Shapes):

A hollow cylindrical pipe has an outer radius $R = 8$ cm, inner radius $r = 6$ cm, and height $h = 14$ cm.

Using $\pi = \frac{22}{7}$ and the formula for the volume of a hollow cylinder $V = \pi(R^2 - r^2)h$:

(a) Calculate the volume of metal used to manufacture the pipe. *(2 marks)*

(b) Find the total surface area of the open hollow pipe using $TSA = 2\pi Rh + 2\pi rh + 2\pi(R^2 - r^2)$. *(1 mark)*

SECTION C: LONG ANSWER & MULTI-FORMULA PROBLEMS

(2 questions $\times$ 5 marks = 10 Marks)

Q9. Multi-Shape Mensuration & Cost Computation:

A composite storage silo is made of a right circular cylinder topped with a closed conical/hemispherical dome. For a cylindrical tank connected to a hemispherical top:

- Radius of cylinder and hemisphere, $r = 7$ m
- Height of the cylindrical portion, $h = 15$ m

Using $\pi = \frac{22}{7}$:

(a) Calculate the total internal volume of the silo using $V_{\text{total}} = V_{\text{cylinder}} + V_{\text{hemisphere}} = \pi r^2 h + \frac{2}{3}\pi r^3$. *(2.5 marks)*

(b) Calculate the total internal surface area available for whitewashing (curved surface of cylinder + curved surface of hemisphere) using $S_{\text{total}} = 2\pi rh + 2\pi r^2$. *(1.5 marks)*

(c) Find the cost of painting this internal surface area at the rate of Rs. 45 per m^2. *(1 mark)*

Q10. Advanced Financial Mathematics & Compound Interest Differentials:

A principal amount of $P =$ Rs. $80{,}000$ is invested for a time period of $t = 1.5$ years at an annual interest rate of $R = 10\%$ per annum.

(a) Calculate the final compound amount A_1 and Compound Interest CI_1 if the interest is **compounded annually** using $A = P\left(1 + \frac{R}{100}\right)^n$. *(2 marks)*

(b) Calculate the final compound amount A_2 and Compound Interest CI_2 if the interest is **compounded half-yearly** using $A = P\left(1 + \frac{R/2}{100}\right)^{2n}$. *(2 marks)*

(c) Compute the exact difference in interest earned between half-yearly and annual compounding methods $(CI_2 - CI_1)$. *(1 mark)*